disadvantage compared to silicon, being closer to insulators than conductors. However, their wide bandgap allows for much thinner dice for the same voltage rating, reducing on-resistance and lowering thermal resistance. Additionally, WBG materials start with lower leakage currents that rise more slowly with temperature increases. These materials can also tolerate higher operating temperatures, enhancing their power handling capability for a given device size despite the limitations of their die attached solder or packaging.

What makes Silicon Carbide (SiC) a Game-Changer in the Semiconductor Industry?

The rising affordability of EVs is underscored by the lower charging costs at public stations, typically ranging from ₹22.20 to ₹31.07 per kWh—significantly cheaper than the expenses associated with diesel-powered vehicles. Over the lifecycle of an EV, this price difference translates into considerable savings. But how substantial is the environmental impact? According to the Association of German Engineers (VDI), a medium-sized EV becomes environmentally beneficial after 90,000km, boasting emissions about 55% lower than traditional combustion engines over a span of 200,000km.

Amidst these developments, SiC is revolutionising power semiconductor technology. Components by various companies highlight SiC's superior performance, showcasing up to 90% lower switching losses compared to silicon alternatives. But why does this matter? This efficiency permits higher switching frequencies, drastically reducing the size of components such as coils and transformers, thus enhancing the compactness and efficiency of the systems.

With the growing demand for electric mobility and sustainable energy solutions, SiC's contribution to enhancing the efficiency and performance of high-voltage power systems becomes more vital. Its robustness in handling the rigorous demands of modern energy needs cements its position as a pivotal material in steering us towards a greener, more efficient future.

Enhanced charging technology with SiC

As these silicon-based semiconductors take the stage, they are set not only to meet current industry needs but also to anticipate future demands, making them pivotal in the evolution of technology across various industries. In a remarkable advancement for electric vehicle technology, the focus has shifted towards enhancing the efficiency and capacity of onboard chargers, heralding a new era of high-performance electric mobility. SiC-based power devices are leading this transformation, offering a suite of benefits that propel the capabilities of electric vehicles (EVs) forward.

With EVs consuming between 11kWh and 30kWh per 100 kilometre—often estimated at 15kWh for typical usage—the surge in electricity demand is undeniable. This growing appetite for power necessitates not only robust energy systems but also innovative solutions like solar carports. These installations, which can be found at corporate parking lots, supermarkets, and residential buildings, not only provide power but also protect vehicles from environmental elements. Additionally, their integration with bidirectional charging systems supports energy storage and grid balancing, enhancing the sustainability of EV ecosystems.

As the EV market continues to expand, the demand for more efficient and faster charging solutions becomes critical. Silicon carbide technology is at the forefront of meeting these needs, offering higher efficiency and reduced power loss compared to traditional silicon-based devices. This shift is not merely incremental; it is pivotal in enhancing the overall performance of electric vehicles. Silicon carbide-based devices provide substantial improvements over their silicon counterparts:

Higher efficiency. SiC devices operate with significantly higher efficiency. This capability leads to lower power loss during conventional procedures, which translates into better overall vehicle performance.

Improved thermal management. Due to its superior thermal conductivity, silicon carbide can handle higher temperatures and improve the reliability and efficiency of chargers.

Greater power density. The ability of SiC to operate at higher temperatures also allows for higher power density within chargers. This means that chargers can be more compact and lightweight, aligning with the current trend toward minimising the size and weight of electronic components in vehicles.

Enhanced switching frequency. SiC devices can switch on and off at much higher frequencies than silicon devices, minimising energy loss during switching. This feature is critical for optimising the efficiency of the energy conversion process in EVs.

Higher operating voltages. Capable of withstanding higher voltages, SiC devices enable the design of more flexible and robust electrical systems within vehicles.

SiC MOSFET: A sustainable future

A specific example of this innovative technology in action is the use of SiC MOSFETs in EV onboard chargers. These devices demonstrate significantly reduced conduction and switching losses, enhancing power efficiency and reducing heat generation during operation. The Kelvin source pin, a feature of some